

Vector Databases and Retrieval Systems for Large-Scale AI Applications: A Survey

Vector databases and retrieval systems have become core infrastructure for large-scale AI applications. They store embedding representations, execute nearest-neighbor and hybrid search, support filtering and reranking, and increasingly serve retrieval-augmented generation, agents, recommendation, multi-modal search, and enterprise knowledge workloads. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run. It connects classical and modern approximate nearest-neighbor indexing with dense neural retrieval, hybrid search, large-scale serving, benchmark methodology, and reliability concerns. The central argument is that vector databases are not merely storage engines for embeddings; they are adaptive retrieval systems whose performance depends on representation learning, index design, update policy, filtering semantics, hardware layout, and application-level quality evaluation.

Keywords:

vector databases; similarity search; approximate nearest neighbors; dense retrieval; hybrid retrieval; retrieval-augmented generation; large-scale AI systems

1 Introduction

Large-scale AI applications increasingly depend on retrieval infrastructure. Search is no longer limited to keyword matching over documents: embeddings produced by neural encoders represent text, images, code, users, items, molecules, and interaction histories in high-dimensional vector spaces. Vector databases and retrieval systems index these vectors, retrieve nearest neighbors, combine vector scores with filters or sparse signals, and return evidence to downstream applications such as question answering, recommendation, semantic search, multi-modal retrieval, and retrieval-augmented generation [9, 15, 112, 156, 206, 221, 233, 297].

The field sits at the intersection of database systems, information retrieval, machine learning, and systems engineering. Approximate nearest-neighbor indexing provides the algorithmic substrate for fast top-k search. Dense retrieval learns the embedding space and determines what neighborhood structure means. Hybrid retrieval and reranking compensate for the limitations of any single representation. Large-scale serving turns these components into con-

tinuously updated systems with latency, throughput, memory, privacy, and freshness constraints [9, 15, 97, 112, 133, 156, 221, 297].

This survey reviews vector databases and retrieval systems as end-to-end AI infrastructure. We emphasize how retrieval quality, index design, system architecture, and application behavior interact.

2 Background

2.1 From Similarity Search to Vector Databases

Similarity search asks for items whose vectors are close to a query vector under a metric or score such as Euclidean distance, cosine similarity, or inner product. Exact search is simple but expensive at scale. Approximate nearest-neighbor methods trade a controlled loss in recall for large gains in speed and memory efficiency. Vector databases wrap these indexes with persistence, update, filtering, metadata, access control, observability, and serving interfaces [9, 15, 97, 112, 133, 156, 221, 297].

2.2 Approximate Nearest-Neighbor Indexing

ANN indexes can be organized around quantization, graph traversal, hashing, trees, inverted files, or hybrids. Product quantization compresses vectors and enables fast distance approximation. Graph methods navigate neighbor links from entry points to promising regions of the space. Disk-backed and GPU-oriented systems redesign layout around memory hierarchy, I/O, and parallelism. The key trade-off is not a single speed-recall curve; it also includes build time, update cost, memory footprint, filter support, tail latency, and hardware efficiency [9, 15, 133, 156, 206, 221, 233, 297].

2.3 Dense Neural Retrieval

Dense retrieval uses neural encoders to map queries and documents into the same vector space. Dual encoders make retrieval efficient because documents can be pre-encoded, while cross-encoders and late-interaction models improve matching by comparing richer token-level representations. Representation learning determines whether vector search returns semantically useful neighbors; the database can only expose the geometry that the encoder creates [56, 111, 118, 141, 154, 167, 247, 284].

2.4 Hybrid Search and Reranking

Practical retrieval often combines dense vectors with sparse lexical signals, metadata filters, business rules, or learned rerankers. Hybrid systems improve robustness for rare terms, identifiers, names, code symbols, numerical constraints, and queries where embeddings over-generalize. Reranking shifts some computation from candidate generation to higher-quality scoring, creating a pipeline in which recall at the vector-search stage and precision at the reranking stage must be evaluated together [9, 97, 112, 167, 183, 206, 221, 255].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with vector databases, vector search, semantic search, similarity search, approximate nearest-neighbor indexing, maximum inner product search, HNSW, product quantization, inverted files, graph indexes, DiskANN, FAISS,

ScaNN, dense retrieval, neural retrieval, dense passage retrieval, dual encoders, late interaction, hybrid retrieval, BM25, reranking, cross-encoders, retrieval-augmented generation, large language models, distributed retrieval, GPU search, benchmark suites, privacy, robustness, and freshness. Records were retained when title or abstract evidence connected them to vector retrieval, ANN indexing, dense or hybrid retrieval, scalable search systems, RAG-oriented retrieval, evaluation, or reliability. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along seven axes. The first is *representation*: raw vectors, compressed codes, learned embeddings, sparse features, or late-interaction token vectors. The second is *index structure*: quantization, graph search, inverted files, hashing, tree structures, or disk/GPU layouts. The third is *retrieval pipeline*: candidate generation, filtering, fusion, reranking, evidence packaging, and feedback. The fourth is *system property*: persistence, dynamic updates, replication, sharding, concurrency, observability, and access control. The fifth is *application*: semantic search, recommendation, RAG, agents, multimodal search, code search, or scientific discovery. The sixth is *evaluation*: recall, precision, latency, memory, freshness, fairness, and downstream task quality. The seventh is *risk*: privacy leakage, poisoning, stale evidence, access control errors, and retrieval-induced hallucination.

3.3 Index Design

Index design begins with the score function and workload. Inner-product search is common for learned retrieval, cosine similarity is common for normalized embeddings, and Euclidean distance appears in many ANN benchmarks. Graph indexes often provide strong recall-latency trade-offs but require memory and careful construction. Quantization improves memory and bandwidth but can degrade ranking. Disk-backed indexes expand capacity but make prefetching, locality, and caching central. Dynamic workloads require insertions, deletions, and compaction without rebuilding the entire system [9, 15, 133, 156, 206, 221, 233, 297].

3.4 Embedding and Retriever Training

The quality of vector search depends on the embedding model. Contrastive learning, hard-negative mining, instruction tuning, distillation, multilingual

training, and domain adaptation shape the neighborhoods that an index will expose. Retriever training must also consider chunking, document structure, multimodal signals, and whether downstream systems need high recall, calibrated scores, diverse evidence, or faithful citations [56, 111, 118, 141, 154, 167, 247, 284].

3.5 Hybrid Retrieval Pipelines

Hybrid retrieval systems combine complementary signals. Lexical retrieval is strong for exact terms and rare entities, dense retrieval is strong for semantic paraphrase, and rerankers can model interactions too expensive for first-stage search. Fusion can happen by score interpolation, rank aggregation, learned routing, or conditional pipelines. For vector databases, hybrid retrieval raises database questions: how filters interact with ANN recall, how metadata predicates are pushed down, and how rerankers are scheduled under latency budgets [9, 97, 112, 167, 183, 206, 221, 255].

3.6 Retrieval-Augmented AI Applications

RAG and agentic applications make retrieval an online reasoning component. A vector database must return evidence that is not only near the query but also useful for generation, attribution, tool use, and user trust. Retrieval affects hallucination, coverage, grounding, and answer freshness. These applications also create new workloads: many short queries, long-context assembly, repeated prefix documents, interactive latency, multi-hop retrieval, and feedback loops from model-generated queries [9, 15, 112, 156, 206, 221, 233, 297].

3.7 Scalable Serving Architecture

Large-scale vector retrieval requires sharding, replication, memory hierarchy management, batching, caching, and hardware-aware layout. GPU indexes can accelerate distance computation but must balance transfer overhead and memory capacity. Disk and SSD indexes support larger corpora but require locality-aware graph traversal or prefetching. Distributed search must merge partial top-k results, handle stragglers, and keep replicas consistent under updates [9, 15, 97, 112, 133, 156, 221, 297].

3.8 Evaluation Methodology

Evaluation should separate index-level and application-level metrics. Index metrics include

recall at k, queries per second, build time, memory footprint, update latency, and tail latency under filters. Retrieval metrics include nDCG, MRR, precision, diversity, and calibration. Application metrics include answer faithfulness, task accuracy, citation support, user satisfaction, and cost. Benchmarks must specify encoders, chunking, filters, hardware, corpus freshness, and reranking because these choices can dominate reported performance [9, 97, 112, 133, 150, 167, 221, 255].

3.9 Reliability, Privacy, and Governance

Retrieval systems expose sensitive information if embeddings leak training data, if access control is enforced after retrieval rather than before it, or if stale and poisoned documents remain highly ranked. Privacy, deletion, freshness, robustness, and provenance therefore belong in the database layer as much as in the application layer. Future systems need auditable retrieval traces, policy-aware indexing, tenant isolation, and quality monitors that detect degraded or unsafe evidence [9, 97, 112, 184, 189, 199, 233, 297].

3.10 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across vector databases, ANN indexing, dense retrieval, hybrid retrieval, RAG applications, scalable serving, evaluation, and reliability.

Vector databases and similarity-search systems

The vector databases and similarity-search systems cluster contains work that develops methods, systems, datasets, or analyses central to this survey [9, 15, 112, 133, 156, 183, 221, 243, 288]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [4, 18, 29, 63, 92, 104, 111, 153, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [33, 44, 125, 195, 228, 239, 251, 284, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [2, 19, 28, 30, 118, 162, 178, 234, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [89, 144, 165, 185, 201, 240, 267, 287, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [66, 77, 132, 192, 215, 225,

[237, 250, 265]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 94, 110, 140, 142, 157, 166, 232, 257]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 35, 52, 177, 209, 230, 236, 241, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 23, 50, 113, 129, 186, 216, 260, 271]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [74, 90, 91, 131, 159, 176, 202, 258, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [59, 68, 101, 102, 108, 231, 270, 292, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [13, 60, 148, 171, 220, 224, 238, 276, 286]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [245].

Approximate nearest neighbor indexing The approximate nearest neighbor indexing cluster contains work that develops methods, systems, datasets, or analyses central to this survey [45, 70, 82, 150, 167, 189, 219, 255, 297]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [53, 114, 141, 146, 248, 254, 262, 281, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 24, 25, 37, 149, 182, 198, 200, 247]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 34, 47, 109, 134, 158, 187, 208, 277]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [81, 120, 121, 143, 145, 188, 204, 244, 253]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [20, 27, 67, 86, 98, 137, 160, 194, 249]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [84].

Dense retrieval and neural information retrieval The dense retrieval and neural information retrieval

cluster contains work that develops methods, systems, datasets, or analyses central to this survey [48, 107, 268].

Hybrid retrieval, filtering, and reranking The hybrid retrieval, filtering, and reranking cluster contains work that develops methods, systems, datasets, or analyses central to this survey [190].

Retrieval-augmented ai applications The retrieval-augmented AI applications cluster contains work that develops methods, systems, datasets, or analyses central to this survey [5, 88, 97, 117, 184, 199, 206, 233, 291]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [76, 130, 161, 169, 197, 212, 213, 242, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [21, 72, 73, 95, 163, 191, 205, 211, 259]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 31, 46, 168, 214, 229, 272, 274, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [61, 96, 128, 164, 179, 207, 227, 246, 263]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [22, 43, 122, 147, 181, 193, 222, 226, 235]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [80, 151, 196].

Scalability, distributed serving, and hardware acceleration The scalability, distributed serving, and hardware acceleration cluster contains work that develops methods, systems, datasets, or analyses central to this survey [38, 58, 64, 69, 78, 106, 174, 256, 280]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [17, 26, 36, 51, 75, 93, 210, 266, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [32, 49, 71, 87, 103, 126, 175, 269, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [39, 100, 173, 223, 252, 279].

Benchmarks and evaluation The benchmarks and evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this

survey [1, 42, 56, 57, 65, 79, 154, 180, 218]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [41, 54, 55, 85, 123, 127, 135, 155, 172]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [62, 99, 115, 124, 136, 138, 152, 217, 275]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [11, 16, 40, 83, 116, 119, 203, 264, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [8, 105, 139, 170, 261, 282].

4 Challenges and Prospects

4.1 Unifying Database Semantics with Learned Representations

Vector databases must combine approximate search with database guarantees. Users expect filtering, deletion, access control, freshness, and observability, while ANN methods often optimize a narrower recall-latency objective. A central research challenge is to make approximation, metadata semantics, and consistency visible and tunable.

4.2 Dynamic and Continually Updated Corpora

AI applications rarely search static corpora. Documents change, embeddings are regenerated, user feedback arrives, and policies require deletion. Dynamic updates stress graph structure, quantization codebooks, cache validity, and benchmark methodology. Incremental indexing and online quality monitoring are therefore essential.

4.3 Evaluation Beyond Recall

High recall does not guarantee useful downstream behavior. A RAG system may need diverse evidence, fresh evidence, policy-compliant evidence, or evidence that supports citations. Future benchmarks should measure full retrieval pipelines and their effects on generated outputs, not only nearest-neighbor recall.

4.4 Hardware-Conscious Retrieval

Vector search is increasingly shaped by memory bandwidth, GPU availability, SSD latency, and network topology. Index algorithms that look attractive

in RAM may behave differently under distributed, filtered, or disk-backed workloads. Hardware-conscious retrieval will require co-design between index structure, compression, scheduling, and placement.

4.5 Trustworthy Retrieval Infrastructure

As retrieval becomes a control point for AI systems, failures become more consequential. Poisoned vectors, stale documents, privacy leakage, and weak tenant isolation can all propagate into model outputs. Trustworthy vector databases will need provenance, permission-aware candidate generation, robust ranking, and auditable evidence trails.

5 Conclusion

Vector databases and retrieval systems are now central infrastructure for large-scale AI. They connect learned representations to operational systems through ANN indexes, dense retrieval models, hybrid search pipelines, scalable serving architectures, and application-level evaluation. This survey reviewed the literature across these layers and argued that progress depends on end-to-end co-design. The next generation of retrieval infrastructure must be fast, accurate, dynamic, policy-aware, and measurable in terms of the AI applications it supports.

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